Tasks

- Derive a linearized state space model of a two-dimensional overheard gantry crane.
- Create a function to evaluate the state space model for given parameters.

Framework provided:

myCraneODE.m	Template for submission
SSmodelParams.mat	Physical parameters for the linear model
ODE_derivation_start.mn	MuPad (Matlab Symbolic Toolbox) script
UsefulCode.m	Some simple code that will prove useful now and in the future
Params_Simscape.mat	Physical parameters for nonlinear Simulink model
SimscapeCrane_StepInput.slx	Step response of the Simulink model

Approach

- Read and understand the derivation of the Euler-Lagrange equations for the pendulum on a cart system provided in the MuPad notebook **ODE_derivation_start.mn**
- Use the MuPad notebook (or otherwise) to derive the matrices (A, B, C, D) of the linearized continuous-time state-space representation. In other words, if the nonlinear system is in the form

$$\frac{\partial}{\partial t}x = \frac{\partial}{\partial t}\begin{bmatrix} X \\ \dot{X} \\ Y \\ \dot{Y} \\ \theta \\ \dot{\theta} \\ \psi \\ \dot{\psi} \end{bmatrix} = F(x,u)$$

then linearize about the equilibrium $(x,u) = (0,0)$, i.e. where $F(0,0) = 0$, so that

$$\frac{\partial}{\partial t}x \approx Ax + Bu.$$

Here $u := [u_x \; u_y]'$ is the input vector, where u_x and u_y are the signals to the motors for the X and Y directions, respectively.

We will assume full state measurements with no direct term, i.e. the measurement vector is given by $y = Cx + Du = x$.

Please try not to confuse the notation for the state vector x and measurements y with the notation for the displacements X and Y in the horizontal X and Y directions. θ and ψ are the angles from the vertical in the X and Y directions, respectively.

Hint: In order to successfully derive A and B for a linearization about zero, you might find it useful to recall that $\sin(\varepsilon) \approx \varepsilon$ and $\cos(\varepsilon) \approx 1$ for $\varepsilon \approx 0$. You might also want to revise material on Jacobian linearization, which is based on Taylor series expansions.

- Use the template **myCraneODE.m** to write a function that evaluates the matrices of the continuous-time or discrete-time model of a gantry crane for given parameters. See the template for the function protocol and conventions.
- You should also do the following, as it will help with the final assignment:
  - Familiarize yourself with **usefulCode.m**. See if you can plot step responses for your linear gantry crane model.
  - Load the parameters from **Params_StepInput.mat**, set the simulation time `T` and sample period `Ts` and run the Simulink model **SimscapeCrane_StepInput.slx**. Observe the animation in the Mechanics Explorer. Observe the outputs using the Simulink Data Inspector. Familiarize yourself with the Simulink environment.
  - Use **lsim** plot the system behaviour of your linear gantry crane model when the input is $\sin(t)$ or similar.

Submission

filename	Description.
myCraneODE.m	Evaluates continuous-time or discrete-time model matrices for given parameters

Details of submission: Edit or upload the Matlab template for the above file into Cody Coursework. You should have received an email invitation to join the Cody Coursework page; if not, please check that you are registered on Blackboard for the course and email me so that we can enrol you.

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- Help
- Course Details expand all
- Mathematical Model of a Gantry Crane
  - State space representation
  - Problem Explore
- Receding Horizon Control
  - Prediction Matrices
  - Cost Function Matrices
  - Linear RHC Law Without Constraints
  - Problem Explore
- RHC with Hard Constraints
  - Stage Constraints
  - Trajectory Constraints
  - QP Constraint Matrices
  - Receding Horizon Controller
  - Problem Explore
- RHC with Soft Constraints
  - Augmenting QP Matrices and Vectors
  - Problem Explore
- Assignment

State space representation

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Test Results

Submitted on 25 Jan 2017 by Kassem El Cheikh Ali | Solution ID: 3311939


```
function [A,B,C,D] = myCraneODE(m,M,MR,r,g,Tx,Ty,Vm,Ts)
% Inputs:
% m = Pendulum mass (kg)
% M = Cart mass (kg)
% MR = Rail mass (kg)
% r = String length (m)
% g = gravitational acceleration (m*s^-2)
% Tx = Damping coefficient in X direction (N*s*m^-1)
% Ty = Damping coefficient in Y direction (N*s*m^-1)
% Vm = Input multiplier (scaler)
% Ts = Sample time of the discrete-time system (s)
% Outputs:
% A,B,C,D = State Space matrices of a discrete-time or continuous-time state space model

% The motors in use on the gantry crane are identical and therefore Vx=Vy=Vm.
Vx=Vm;
Vy=Vm;

% replace A,B,C,D with the correct values
A = [[0,1, 0, 0, 0, 0, 0, 0]; [0,Tx/(m*r*(1/r - (M + MR + m)/(m*r))), 0, 0, -g/(r*(1/r - (M + MR + m)/(m*r))), 0, 0, 0]; [0,0, 0
B = [[0, 0]; [-Vx/(m*r*(1/r - (M + MR + m)/(m*r))), 0]; [0, 0]; [0, -Vy/(m*r*(1/r - (M + m)/(m*r)))]; [0, 0]; [Vx/(m*r^2*(1/r - (M
C = eye(8);
D = zeros(8,2);
% if Ts>0 then sample the model, otherwise return a continuous-time model
% with a zero-order hold (piecewise constant) input
if Ts>0
    css = ss(A,B,C,D);
    dss = c2d(css,Ts,'zoh');
    A = dss.A;
    B = dss.B;
    C = dss.C;
    D = dss.D;
    % Return discrete-time SS model matrices
end

end
```

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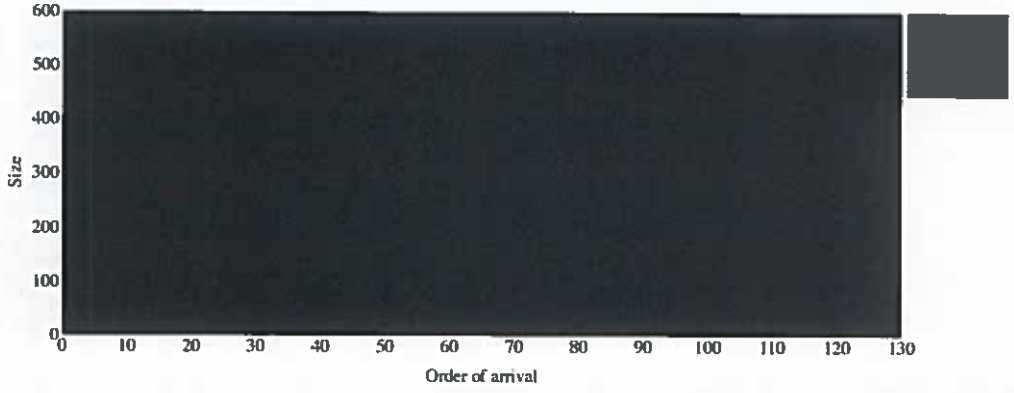
State space representation

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Test Results

Submitted on 23 Jan 2017 by CAILONG HUA | Solution ID: 3275101

```
function [A,B,C,D] = myCraneODE(m,M,MR,r,g,Tx,Ty,Vm,Ts)
% Inputs:
% m = Pendulum mass (kg)
% M = Cart mass (kg)
% MR = Rail mass (kg)
% r = String length (m)
% g = gravitational acceleration (m*s^-2)
% Tx = Damping coefficient in X direction (N*s*m^-1)
% Ty = Damping coefficient in Y direction (N*s*m^-1)
% Vm = Input multiplier (scalar)
% Ts = Sample time of the discrete-time system (s)
% Outputs:
% A,B,C,D = State Space matrices of a discrete-time or continuous-time state space model

% The motors in use on the gantry crane are identical and therefore Vx=Vy=Vm.
Vx=Vm;
Vy=Vm;

% replace A,B,C,D with the correct values
A=zeros(8,8);
A(:,1)=[0 0 0 0 0 0 0 0];
A(:,2)=[1 -Tx/(M+MR) 0 0 0 Tx/(r*(M+MR)) 0 0];
A(:,3)=[0 0 0 0 0 0 0 0];
A(:,4)=[0 0 1 -Ty/M 0 0 0 Ty/(M*r)];
A(:,5)=[0 (g*m)/(M+MR) 0 0 0 -g*(M+MR+m)/(r*(M+MR)) 0 0];
A(:,6)=[0 0 0 0 1 0 0 0];
A(:,7)=[0 0 0 g*m/M 0 0 0 -g*(M+m)/(M*r)];
A(:,8)=[0 0 0 0 0 0 1 0];
B=zeros(8,2);
B(:,1)=[0 Vx/(M+MR) 0 0 0 -Vx/(r*(M+MR)) 0 0];
B(:,2)=[0 0 0 Vy/M 0 0 0 -Vy/(M*r)];
C=eye(8);
D=zeros(8,2);
sys=ss(A,B,C,D);
% if Ts>0 then sample the model with a zero-order hold (piecewise constant)
% input, otherwise return a continuous-time model
if Ts>0
%Return discrete-time SS model matrices
    sys_d=c2d(sys,Ts);
    A=sys_d.A;
    B=sys_d.B;
    C=sys_d.C;
    D=sys_d.D;
else
```

```
    A=sys.A;
    B=sys.B;
    C=sys.C;
    D=sys.D;

end

end
```

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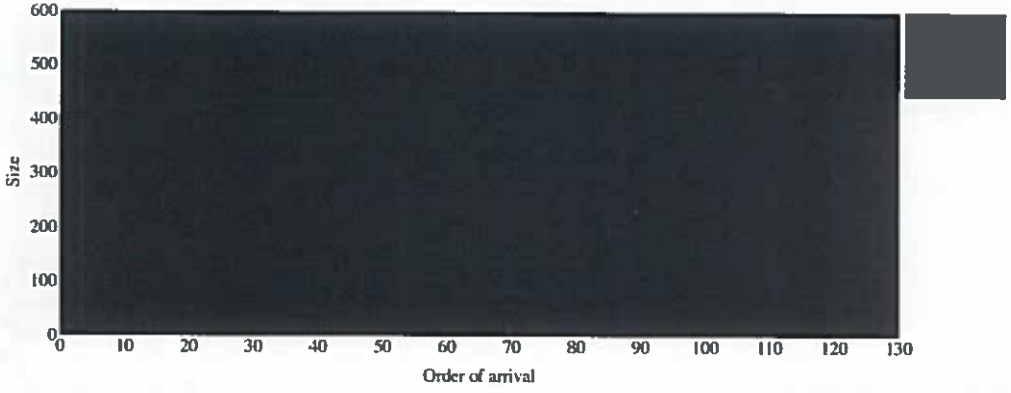
State space representation

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Test Results

Submitted on 23 Jan 2017 by Yutong Ji | Solution ID: 3271642

```
function [A,B,C,D] = myCraneODE(m,M,MR,r,g,Tx,Ty,Vm,Ts)
% Inputs:
% m = Pendulum mass (kg)
% M = Cart mass (kg)
% MR = Rail mass (kg)
% r = String length (m)
% g = gravitational acceleration (m*s^-2)
% Tx = Damping coefficient in X direction (N*s*m^-1)
% Ty = Damping coefficient in Y direction (N*s*m^-1)
% Vm = Input multiplier (scalar)
% Ts = Sample time of the discrete-time system (s)
% Outputs:
% A,B,C,D = State Space matrices of a discrete-time or continuous-time state space model

% The motors in use on the gantry crane are identical and therefore Vx=Vy=Vm.
Vx=Vm;
Vy=Vm;

% replace A,B,C,D with the correct values
A=[0,1,0,0,0,0,0,0;
    0,-Tx/(M+MR),0,0,m*g/(M+MR),0,0,0;
    0,0,0,1,0,0,0,0;
    0,0,0,-Ty/M,0,0,m*g/M,0;
    0,0,0,0,0,1,0,0;
     0,Tx/(M*r+MR*r),0,0,-m*g/(M*r+MR*r)-g/r,0,0,0;
    0,0,0,0,0,0,0,1;
    0,0,0,Ty/(M*r),0,0,-m*g/(M*r)-g/r,0;
    ];
B=[0,0;
    Vx/(M+MR),0;
    0,0;
    0,Vy/(M);
    0,0;
    -Vx/(M*r+MR*r),0;
    0,0;
    0,-Vy/(M*r);
  ];
C=eye(8);
D=zeros(8,2);
% if Ts>0 then sample the model with a zero-order hold (piecewise constant) input
% otherwise return a continuous-time model
if Ts>0
%     Return discrete-time SS model matrices
Dsys=c2d(ss(A,B,C,D),Ts,'zoh');
```

```
[A,B,C,D]=ssdata(Dsys);
end

end
```

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